

This is a conceptual ownership diagram, not a literal MCU address map. Exact flash and RAM addresses depend on the target, linker script, Espressif build, and enabled services.

The companion [memory lifetime diagram](#) shows when these objects and temporary regions are created, used, retained, and released.



- The compiled arena contains native records, but its physical storage is an Esprimo flat string in the GC-managed JsVar pool.
- The machine owns the arena and retained-value container through hidden, garbage-collector-visible children. Releasing the last machine or actor reference makes that graph eligible for collection.
- Actors retain their shared compiled machine but separately own all changing runtime state. Actor A and Actor B therefore share the arena without sharing active state, context, snapshots, faults, subscriptions, or operation state.
- Native records refer to JavaScript values by retained-slot index. They never return a `JsVar *`, `JsVarRef`, or pointer into the arena between calls.
- Compiler workspace, coordinator state, locks, pending values, and callback frames are temporary. Every success and failure path must release them before returning to application JavaScript.
- XFSM has no persistent native-heap allocation or global actor registry. Persistent state remains reachable through ordinary Esprimo object graphs.
- Retaining a flash-backed callback keeps its JavaScript function value reachable but does not copy its source into RAM. Its Storage or module backing must remain valid while the machine can invoke it.

- `save()` records a stable interpreter object graph only after Espruino returns to idle. It does not persist a live coordinator frame, arena pointer, pending state, or partially completed notification.